

Video Transcript

Video 8: Using NiFi to Create an ETL Pipeline (09:24)

It is now time to bring it all together, NiFi, MySQL, and ETL. As you can see here, this is the architecture that we have built. We created a network; we then created a container with a database server and then we added a platform that allows us to do ETL. Now we're going to be working within that platform, NiFi, to be able to source that data, extract, transform, and view within that environment, that would be the equivalent of loading. We're going to start off by creating the database connection pool, and we're going to do that by creating or clicking on the gear icon there, as you can see. Then, we're going to go ahead and select 'CONTROLLER SERVICES', and then within that page, we're going to click on the plus. Now from the popup, we're going to go ahead and select the connection pool, and the option for that is DBCPConnectionpool. As you can see here, it's the second option. We're going to go ahead and add that. Next, we're going to go ahead and configure that. We'll start off by clicking on the gear icon here.

The very first thing that we will do is that we will call this MySQL. We'll then go ahead and move to properties, and the very first property that we're going to set is the connection URL. And I'm going to go ahead and set there the general pattern of what that connection URL is, notice that it's JDBC MySQL and note here that my path to the location, the URL, is mysqllabel, that is what I called my container. And, the internal networking, the network that we are creating, that we created when we were setting up our containers will know how to do that resolution. If we hadn't done that, we would have had to discover what the IP was within our host machine that, that specific container was running under, and that would be fine as well.

However, that makes it more brittle because you have to do discovery every single time you run your network, you'll get new IPs when you delete containers. And so, in this case, doing this makes it much easier for us. And so, that's the advantage of being able to create a container, to create a network and being able to assign containers to that network. So, I'm going to go ahead and click 'OK'. Next, I'm going to set the database driver class name. And as you can see there, that is simply what the convention is for this type of driver.

So, I'm going to go ahead and click 'OK'. Next, we're going to go ahead and set the database driver location. When we created our container, our NiFi container, and we uploaded a driver onto a directory that is the path that I'm going to be using. So, '/opt/nifi/drivers' is where I uploaded that container that we downloaded from mysql.com, that is the location. And so, in this case, this is what we are adding. Now, I'm simply going to end that with the name of the file and click 'OK'.

And I'm now going to go ahead and create my database user, and this is simply root. And I'm going to set the password, and the password is the one I have been using my new pass. Needless to say, that this is something you should be careful with. Right now, we're simply doing a demonstration. I will delete this container when I am done. You should do the same, they should not be up and available because it is a security threat. And here, of course, I'm referring to recording and distributing this openly with the credentials that you're using.

So, let's go ahead and apply. And now, we're going to go ahead and activate. There is a lightning bolt next to the gear that says enable. And when I click on this, I'm going to go ahead and choose the option here for service and referencing components. Then I'm going to go ahead and select 'ENABLE'. And you can see there that that is taking place and then have the check marks there and that we have a blue lightning bolt saying enabled.

And so, now that we're finished configuring the database connection pool, we'll go ahead and add a couple of processors. The very first one, and you can do so by simply dragging and dropping, will be the 'ExecuteSQL,' and this one will be the one that goes out and queries the database and brings back some SQL. So, you can see here that one, I'm going to go ahead and add it, and you can see there that we get a nice widget within the screen. Then I'm going to go ahead and bring in another processor and this one, we're going to select log attribute. As you can see there, we get that option. I'm going to go ahead and add it onto the screen.

I'm now going to go ahead and connect both of them as you can see there. And I'm going to go ahead and select for relationships, success. Now, we're going to go ahead and configure our SQL. We'll start off here by setting the option to automatically turn when we have a failure. Next, we're going to go ahead and move to scheduling, and we're going to go ahead and run this every 30 second. We're then going to go ahead and move to properties, and we're going to go ahead and set the database connection pooling, and that's what we configured initially.

And as you can see there, we get that option, that is the name that we set. Now, we're going to

go ahead and enter our query. And the query that I've entered here is the same one that we entered when we were working on the Workbench, and you can see here that I am specifying my database in addition to the table. So, I'm going to go ahead and select 'OK'. Then I will enter apply, and I will go ahead and start my processor. And as you can see there now, we have an item on the queue.

If we go ahead and open this up, and look at settings, this is you can see there the information for this this intermediate step. And if we right click and take a look here at the queue, we can take a look at the document and we can look here at the first one. And if we take a look at the formatted form of it, you can see here that we are reading the data that we put into the database. Now you can see there that those records are accumulating as well. So, I'm going to go ahead and close and I'm going to go ahead and stop. Now as you saw, this is a pretty graphical type of interface, most of what you were doing was configuration, clicking buttons, choosing options as opposed to expressing all of that in code.

So, this would be an environment which you could call low-code or no-code. Now, there's still a great deal of knowledge that you need to have to be able to choose the configurations, but this is a very visual approach as opposed to a very code-driven approach. Now you can see there that even in the minimal example that we did, we were pulling some data, we were creating a processor, we will then, that processor, then was creating FlowFiles, and we had a connector through which we could take a look at that queue and view the data that was being pulled.

Now, you can imagine that in the same manner, you could go ahead and continue this pipeline to be able to include many other steps that did additional transformations. In this case, we were simply pulling the data, viewing it in a different format, and then confirming that we were getting the data that we expected, that MySQL database could continue to populate. You could continue to pull that information. So, considering everything that is being done, this is a tremendous platform that greatly simplifies being able to do ETL, extract, transform, and load.